

Titration report



Method

Method ID	CDC_GranTED-DYN	Release state	Released
Name	Titration with HCl - Gran Verification	Modified by	PredefinedUser
Modified on	2/5/2026 6:15:50 AM		

Summary

Samples

Sample ID	Name	Result	State
1	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.007599 mL
		Concentration R2	0.10015 mol/L
2	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.005214 mL
		Concentration R2	0.10010 mol/L
3	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.041297 mL
		Concentration R2	0.10083 mol/L

Statistics

R1

Mean	5.018037 mL	Minimum	5.005214 mL
Standard deviation	0.020179 mL	Rel. standard deviation	0.402 %
Total samples	3	Samples excluded	0

R2

Mean	0.10036 mol/L	Minimum	0.10010 mol/L
Standard deviation	0.00041 mol/L	Rel. standard deviation	0.406 %
Total samples	3	Samples excluded	0

Main Results

Sample ID1	Sample ID2	Result Title	Name	Result	State
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.007599 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10015 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.005214 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10010 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.041297 mL	OK

Titration report

LabX

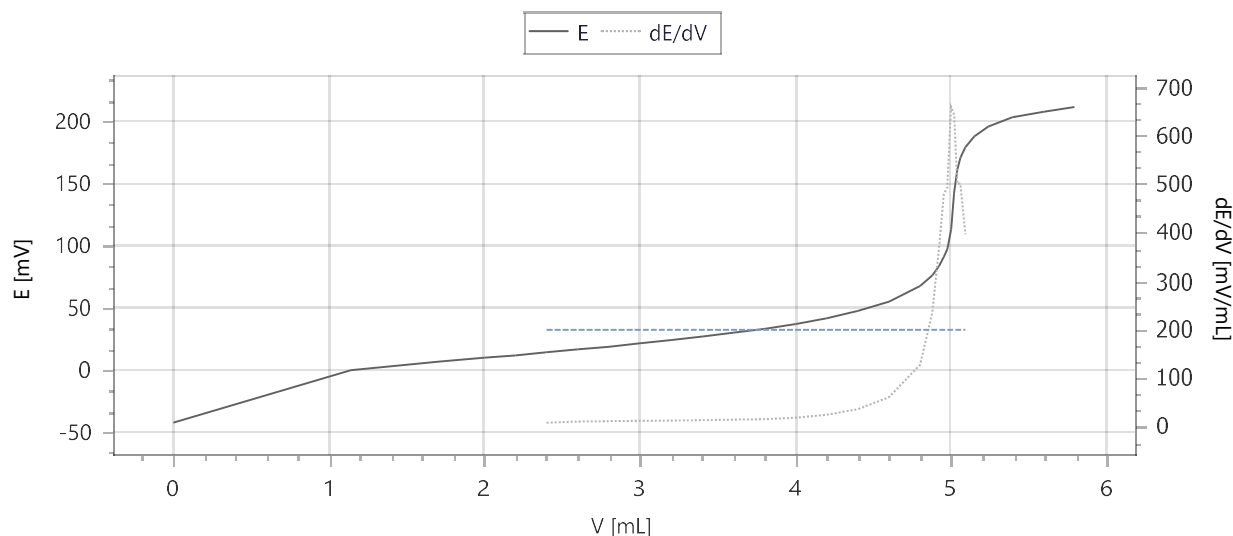
0.1 M TRIS - 5 mL
in 60 mL H₂O

Concentration R2

0.10083 mol/L OK

Titration (EQP) [1] Scope 1/1, Sample 1/3

E - V & dE/dV - V Curve Scope 1/1, Sample 1/3



Measured Values Scope 1/1, Sample 1/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-42.5		25.0			
9	1.1430	-0.1	1.1430	25.0	42.4		
17	1.7145	6.7	0.5715	25.0	6.8		
22	2.0000	10.0	0.2855	25.0	3.3		
58	2.2000	11.7	0.2000	25.0	1.7		
62	2.4000	14.3	0.2000	25.0	2.6	9.12	
66	2.6000	16.5	0.2000	25.0	2.2	11.60	
72	2.8000	18.6	0.2000	25.0	2.1	12.34	
76	3.0000	21.4	0.2000	25.0	2.8	13.07	
79	3.2000	24.2	0.2000	25.0	2.8	13.56	
82	3.4000	27.0	0.2000	25.0	2.8	14.38	
87	3.6000	29.9	0.2000	25.0	2.9	15.49	
92	3.8000	33.2	0.2000	25.0	3.3	16.42	
96	4.0000	37.0	0.2000	25.0	3.8	19.55	
102	4.2000	41.5	0.2000	25.0	4.5	25.48	
105	4.4000	47.5	0.2000	25.0	6.0	37.02	
110	4.6000	54.9	0.2000	25.0	7.4	61.62	
118	4.8000	67.7	0.2000	25.0	12.8	128.27	
127	4.8785	76.1	0.0785	25.0	8.4	235.37	
135	4.9225	83.6	0.0440	25.0	7.5	371.01	
143	4.9525	91.0	0.0300	25.0	7.4	479.52	
150	4.9745	97.3	0.0220	25.0	6.3	492.15	
161	4.9990	113.2	0.0245	25.0	15.9	658.53	
	5.007599	125.7				658.53	EQP
173	5.0190	142.3	0.0200	25.0	29.1	642.31	

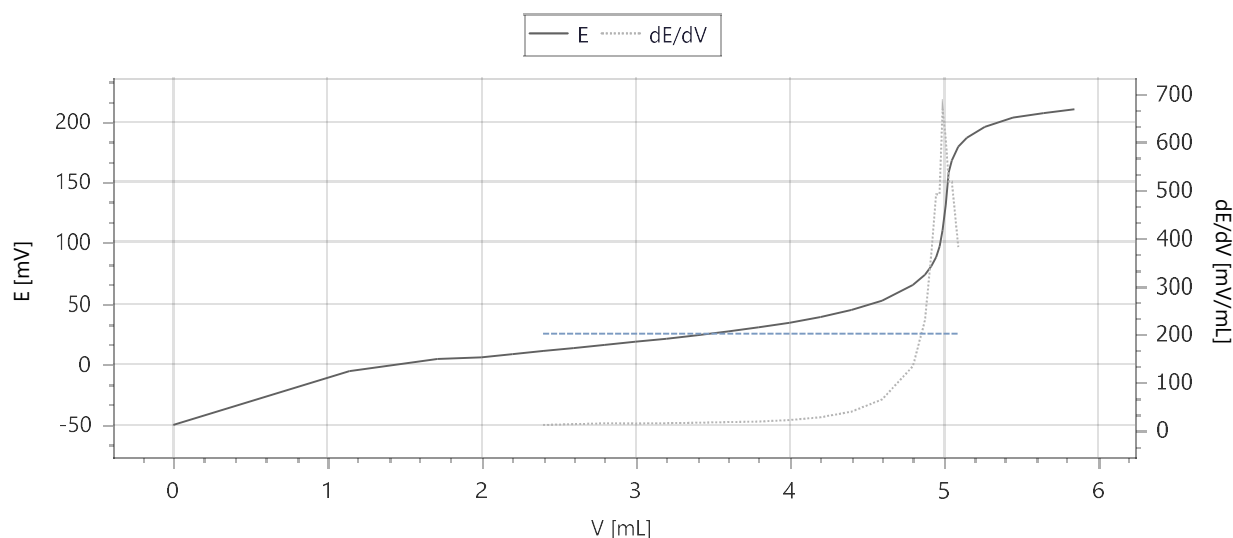
Titration report

LabX

177	5.0390	161.2	0.0200	25.0	18.9	500.72
182	5.0590	171.0	0.0200	25.0	9.8	500.90
186	5.0910	179.3	0.0320	25.0	8.3	396.65
189	5.1480	188.0	0.0570	25.0	8.7	
192	5.2385	196.1	0.0905	25.0	8.1	
195	5.3925	203.6	0.1540	25.0	7.5	
203	5.5925	208.0	0.2000	25.0	4.4	
206	5.7925	211.9	0.2000	25.0	3.9	

Titration (EQP) [1] Scope 1/1, Sample 2/3

E - V & dE/dV - V Curve Scope 1/1, Sample 2/3



Measured Values Scope 1/1, Sample 2/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-50.7		25.0			
9	1.1430	-6.0	1.1430	25.0	44.7		
13	1.7145	3.9	0.5715	25.0	9.9		
18	2.0000	5.2	0.2855	25.0	1.3		
51	2.2000	8.0	0.2000	25.0	2.8		
54	2.4000	10.6	0.2000	25.0	2.6	9.87	
58	2.6000	13.1	0.2000	25.0	2.5	11.67	
61	2.8000	15.5	0.2000	25.0	2.4	13.34	
65	3.0000	18.2	0.2000	25.0	2.7	12.94	
68	3.2000	20.7	0.2000	25.0	2.5	13.63	
72	3.4000	23.7	0.2000	25.0	3.0	14.55	
76	3.6000	26.7	0.2000	25.0	3.0	15.85	
79	3.8000	30.2	0.2000	25.0	3.5	17.06	
84	4.0000	34.0	0.2000	25.0	3.8	20.16	
87	4.2000	38.7	0.2000	25.0	4.7	25.93	
90	4.4000	44.6	0.2000	25.0	5.9	37.54	
95	4.6000	52.3	0.2000	25.0	7.7	63.05	
103	4.8000	65.1	0.2000	25.0	12.8	132.26	
111	4.8785	73.7	0.0785	25.0	8.6	228.34	

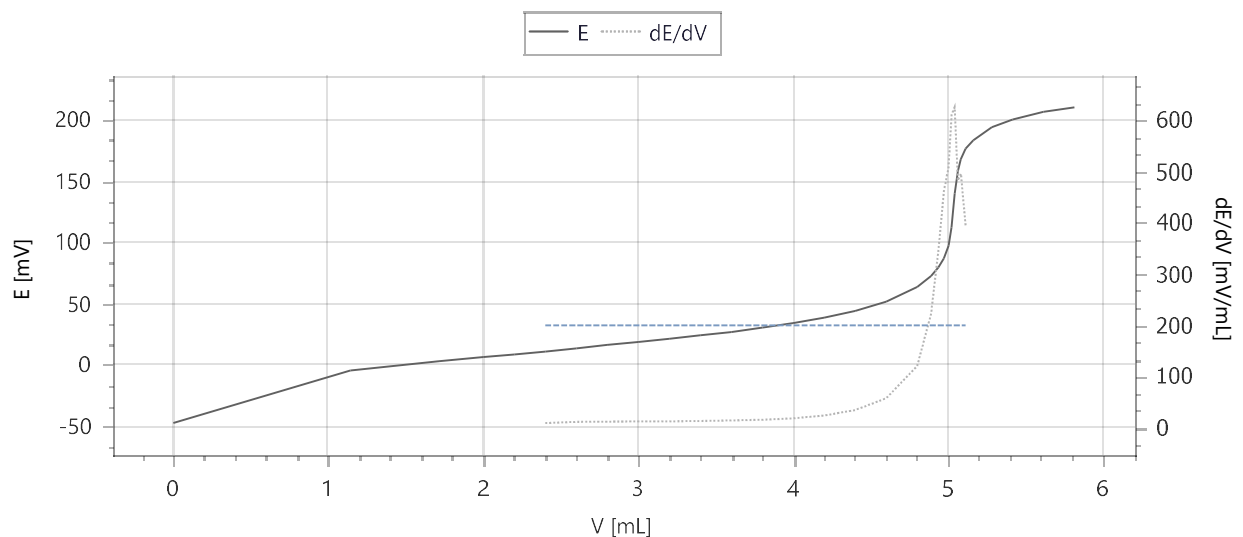
Titration report

LabX

120	4.9205	80.9	0.0420	25.0	7.2	370.04	EQP
127	4.9500	88.2	0.0295	25.0	7.3	489.72	
136	4.9725	97.2	0.0225	25.0	9.0	490.25	
147	4.9925	110.4	0.0200	25.0	13.2	667.22	
	5.005214	123.8				667.79	
150	5.0125	131.5	0.0200	25.0	21.1	607.01	
158	5.0325	158.2	0.0200	25.0	26.7	516.50	
161	5.0525	168.3	0.0200	25.0	10.1	516.31	
165	5.0945	179.6	0.0420	25.0	11.3	380.07	
168	5.1500	187.2	0.0555	25.0	7.6		
171	5.2670	196.2	0.1170	25.0	9.0		
174	5.4485	203.9	0.1815	25.0	7.7		
183	5.6485	207.4	0.2000	25.0	3.5		
187	5.8485	210.8	0.2000	25.0	3.4		

Titration (EQP) [1] Scope 1/1, Sample 3/3

E - V & dE/dV - V Curve Scope 1/1, Sample 3/3



Measured Values Scope 1/1, Sample 3/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-47.2		25.0			
9	1.1430	-4.3	1.1430	25.0	42.9		
15	1.7145	3.3	0.5715	25.0	7.6		
20	2.0000	6.7	0.2855	25.0	3.4		
55	2.2000	8.7	0.2000	25.0	2.0		
60	2.4000	11.1	0.2000	25.0	2.4	9.95	
63	2.6000	13.7	0.2000	25.0	2.6	12.28	
66	2.8000	16.5	0.2000	25.0	2.8	12.79	
69	3.0000	18.9	0.2000	25.0	2.4	13.12	
72	3.2000	21.5	0.2000	25.0	2.6	13.32	
75	3.4000	24.3	0.2000	25.0	2.8	13.97	
78	3.6000	27.1	0.2000	25.0	2.8	15.08	
82	3.8000	30.6	0.2000	25.0	3.5	16.42	

Titration report

LabX

87	4.0000	34.2	0.2000	25.0	3.6	19.19	EQP
91	4.2000	38.7	0.2000	25.0	4.5	24.70	
96	4.4000	44.2	0.2000	25.0	5.5	35.33	
100	4.6000	51.7	0.2000	25.0	7.5	58.22	
107	4.8000	63.7	0.2000	25.0	12.0	121.07	
112	4.8895	72.7	0.0895	25.0	9.0	222.36	
116	4.9365	79.8	0.0470	25.0	7.1	345.30	
122	4.9725	87.1	0.0360	25.0	7.3	461.43	
131	5.0020	97.6	0.0295	25.0	10.5	507.43	
145	5.0220	112.4	0.0200	25.0	14.8	609.91	
	5.041297	139.1				627.23	
158	5.0420	140.1	0.0200	25.0	27.7	624.84	
164	5.0620	158.3	0.0200	25.0	18.2	482.25	
167	5.0820	168.0	0.0200	25.0	9.7	495.52	
170	5.1125	176.6	0.0305	25.0	8.6	393.89	
178	5.1615	183.3	0.0490	25.0	6.7		
181	5.2815	194.2	0.1200	25.0	10.9		
184	5.4150	200.4	0.1335	25.0	6.2		
187	5.6150	206.6	0.2000	25.0	6.2		
190	5.8150	210.4	0.2000	25.0	3.8		

Sample Series

Name	Internal ID
Label 1	Label 2
Label 3	Label 4
Label 5	Label 6

Comment

Sample

Sample ID 1
Sample ID 2
Sample ID 3

Sample Parameters

Type	Name	Qualified name	Value
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Used Instruments

Name	T9_MSG_SA_002	Serial number	B551866924
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Responsible User

Time	Login name	Full name
05.02.2026 06:30:27	PredefinedUser	Administrator